

Apply filters to SQL queries

Project description

In this project, I will use SQL filtering to retrieve information on a variety of login attempts and employees within a database based on different criteria. I will accomplish this using LIKE and logic gates to filter data.

Retrieve after hours failed login attempts

```
MariaDB [organization]> clear
MariaDB [organization]> SELECT * FROM log_in_attempts WHERE login_time > "18:00"
AND success = FALSE;
+-----+-----+-----+-----+-----+-----+-----+
| event_id | username | login_date | login_time | country | ip_address | success |
+-----+-----+-----+-----+-----+-----+-----+
| 2 | apatel | 2022-05-10 | 20:27:27 | CAN | 192.168.205.12 | 0 |
```

Retrieve login attempts on specific dates

```
MariaDB [organization]> SELECT * FROM log_in_attempts WHERE login_date = "2022-05-08"
OR login_date = "2022-05-09";
+-----+-----+-----+-----+-----+-----+-----+
| event_id | username | login_date | login_time | country | ip_address | success |
+-----+-----+-----+-----+-----+-----+-----+
| 1 | jrafael | 2022-05-09 | 04:56:27 | CAN | 192.168.243.140 | 1 |
```

Retrieve login attempts outside of Mexico

```
MariaDB [organization]> SELECT * FROM log_in_attempts WHERE NOT country LIKE "MEX%";
```

event_id	username	login_date	login_time	country	ip_address	success
1	jrafael	2022-05-09	04:56:27	CAN	192.168.243.140	1

Retrieve employees in Marketing

```
MariaDB [organization]> SELECT * FROM employees WHERE department = "Marketing" AND office LIKE "East%";
```

employee_id	device_id	username	department	office
1000	a320b137c219	elarson	Marketing	East-170

Retrieve employees in Finance or Sales

```
MariaDB [organization]> SELECT * FROM employees WHERE department = "Finance" OR department = "Sales";
```

employee_id	device_id	username	department	office
1003	d394e816f943	sgilmore	Finance	South-153
1007	h174i497j413	wjaffrey	Finance	North-406
1008	i858j583k571	abernard	Finance	South-170
1009	NULL	lrodriqu	Sales	South-134

Retrieve all employees not in IT

```
MariaDB [organization]> SELECT * FROM employees WHERE NOT department = "Information Technology";
```

employee_id	device_id	username	department	office
1000	a320b137c219	elarson	Marketing	East-170
1001	b239c825d303	bmoreno	Marketing	Central-276

Summary

Overall, I have used a range of SQL filtering methods in the WHERE line to retrieve different information.